Sophia Simons
HW 4

1) a. [SP]+1 → Q: rac=1, rn=2, ib, P1, oadder

b. [PC] → T2: rac=1, rn=3, it2

c. [PC]+1 → PC: cannot read and write to PC in one cycle?

d. [T1]-[T5] → Q: oa, ot5, ib, oadder, comp, P1

e. [T5]-[T1] → Q: ot5, ib, oa, oadder, comp, P1

f. [T1]-1 → Q: ot1, ib, oadder, comp, P1

g. One's comp. of [AC] → Q: rac=1, rn=0, ib, comp, oadder

h. Two's comp of [T1] → Q: ot1, ib, comp, oadder, P1

i. [T4] → T2, [T1] → X: ot4, it2; ot1, wac=1, wn=1

j. Exchange contents of T2 and Q: ot2, iQ, oQ, it2

k. [[MAR]] → MDR, [T1] → Q, [AC] → T5:
omar, imdr; oa, oadder; rac=1, rn=0, it5

l. [MDR] → IR, [AC] → T5
data transfer cannot occur in one clock cycle?

2)

ADDR	INSTRUCTION	BINARY	HEX
0x00	MOVE #$6,(SP)	0011 0110 0001 0110	0x3616
0x01	value 6 from imm. value		0x006
0x02	JSR SUBR	0000 0111 0101 0000	0x0750
0x03	addr of SUBR		0x00E
0x04	NEG AC	0000 0100 0000 0000	0x0400
0x05	COMP X	0000 0101 0000 0010	0x0501
0x06	CLR -5(AC)	0000 0001 0100 0000	0x0140
0x07	offset from index -5		0xFFFB
0x08	EXG (AC)+, X	0100 0010 0000 0001	0x4201
0x09	ADD -(AC),X	0001 0011 0000 0001	0x1301
0x0A	SUB #$20, DATA	0010 0110 0101 0000	0x2650
0x0B	20 from imm value		0x0020
0x0C	DATA from abs addr		0x0019
0x0D	HALT	0000 0000 0000 0000	0x0000
0x0E	INC SP	0000 0010 0000 0010	0x0202
0x0F	MOVE (SP),X	0011 0001 0000 1001	0x3109
0x10	SUB X, AC	0010 0000 0000 0100	0x2004
0x11	TST AC	0000 1000 0000 0000	0x0800
0x12	BMI SKIP	0000 0000 1001 0001	0x0091
0x13	branch offset		0x0003
0x14	INC AC	0000 0010 0000 0000	0x0200
0x15	JMP LOOP	0000 0110 0101 0000	0x0650
0x16	addr of LOOP		0x0010
0x17	DEC SP	0000 0011 0000 0010	0x0302
0x18	RTS	0000 0000 1000 0000	0x0080
0x19	DATA		0xAB00

Branch offset = 0x17 − (PC_original + 2)
= 0x17 − (0x12 + 2)
= 0x0003

3)

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     0011 AC = 3
     0008 X = 8
     0010 SP = 2
     0000 PC = 0
     0000 CVZN
1    3009 MOVE SP,X
2    3109 MOVE (SP),X
3    3209 MOVE (SP)+,X
4    3309 MOVE -(SP),X
5    3409 MOVE 1(SP),X
     0001
6    3501 MOVE 0x000F,X
     000F
7    3601 MOVE #0xF1F2,X
     F1F2
8    3028 MOVE SP,(AC)+
9    300D MOVE PC,X
10   3630 MOVE #0x0000,-(AC)
     0000
     0000 HALT
     FFFF
     0003
     8000
     0005
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Summary.1:

 At end of instruction 1
   ac      x     sp    pc    cvzn
   11     10     10     1    0000
 At end of instruction 2
   ac      x     sp    pc    cvzn
   11     03     10     2    0000
 At end of instruction 3
   ac      x     sp    pc    cvzn
   11     03     11     3    0000
 At end of instruction 4
   ac      x     sp    pc    cvzn
   11     03     10     4    0000
 At end of instruction 5
   ac      x     sp    pc    cvzn

   11      8     10     6    0001
 At end of instruction 6
   ac      x     sp    pc    cvzn
   11   FFFF     10     8    0001
 At end of instruction 7
   ac      x     sp    pc    cvzn
   11   F1F2     10     A    0001
memory write:: 0010->M(0011)
 At end of instruction 8
   ac      x     sp    pc    cvzn
   12   F1F2     10     B    0000
 At end of instruction 9
   ac      x     sp    pc    cvzn
   12      C     10     C    0000
memory write:: 0000->M(0011)
 At end of instruction 10
   ac      x     sp    pc    cvzn
   11      C     10     E    0010
```